higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

31 July 2025

This marking guideline consists of 11 pages.

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QUESTION 1

1.1 1.1.1

$$y=\lim_{x\to\infty}\frac{x}{\ln^3(2x-1)} \qquad \frac{\infty}{\infty}$$

$$=\lim_{x\to\infty}\frac{2x-1}{6\ln^2(2x-1)} \qquad \frac{\infty}{\infty}$$

$$=\lim_{x\to\infty}\frac{4x-2}{24\ln(2x-1)} \qquad \frac{\infty}{\infty}$$

$$=\lim_{x\to\infty}\frac{8x-4}{48}$$

$$=\infty$$ (4)

1.2

$$\cos x=\lim_{x\to 0}\frac{(\tan^{-1}x)^2}{x} \qquad \frac{0}{0}$$

$$=\lim_{x\to 0}\frac{2\tan^{-1}x\times\frac{1}{1+x^2}}{1}$$

$$=0$$ (2)

[6]

QUESTION 2

2.1 2.1.1

$$f(x+h)$$

$$=8(x+h)^{-\frac{1}{2}}$$

$$=8\left[\frac{x^{-\frac{1}{2}}h^0}{0!}-\frac{\frac{1}{2}x^{-\frac{3}{2}}h}{1!}+\frac{\frac{3}{4}x^{-\frac{5}{2}}h^2}{2!}+\ldots\ldots\right]$$

$$=8x^{-\frac{1}{2}}-4x^{-\frac{3}{2}}h+3x^{-\frac{5}{2}}h^2+\ldots\ldots\ldots$$ (2)

2.1.2

$$f(x+h)-f(x)$$

$$=-4x^{-\frac{3}{2}}h+3x^{-\frac{5}{2}}h^2+\ldots\ldots\ldots$$ (1)

2.1.3

$$\frac{f(x+h)-f(x)}{h}$$

$$=-4x^{-\frac{3}{2}}+3x^{-\frac{5}{2}}h+\ldots\ldots$$ (1)

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2.1.4 $\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h} = -4x^{-\frac{3}{2}}$ (1)

2.2 2.2.1 $y=(5+\log_4 x)^4(7x-e^{\sin 4x})^5$

$\dfrac{dy}{dx}=(7x-e^{\sin 4x})^5\times 4(5+\log_4 x)^3\times\dfrac{1}{x\ln 4}+(5+\log_4 x)^4\times 5(7x-e^{\sin 4x})^4\times(7-e^{}$ (4)

2.2.2 $y=\tan^3\left[\sin\left(\dfrac{7x^2+2}{x-3}\right)\right]$

$\dfrac{dy}{dx}=3\tan^2\left[\sin\left(\dfrac{7x^2+2}{x-3}\right)\right]\times sec^2\left[\sin\left(\dfrac{7x^2+2}{x-3}\right)\right]\times\cos\left(\dfrac{7x^2+2}{x-3}\right)\times\dfrac{7x(x-3)-(7}{(x-3)^2}$ (5)

2.3 $y=(5-4x)^{\frac{1}{x}}$

$\ln y=\dfrac{1}{x}\ln(5-4x)$

$\dfrac{1}{y}\dfrac{dy}{dx}=-\dfrac{1}{x^2}\times\ln(5-4x)+\dfrac{1}{x}\times\dfrac{1}{5-4x}\times(-4)$

$\dfrac{dy}{dx}=y\left[-\dfrac{\ln(5-4x)}{x^2}-\dfrac{4}{x(5-4x)}\right]$

$=(5-4x)^{\frac{1}{x}}\left[-\dfrac{\ln(5-4x)}{x^2}-\dfrac{4}{x(5-4x)}\right]$ (4)

2.4 2.4.1 $x^3+5xy-2y^2=8$

$3x^2+5y+5x\dfrac{dy}{dx}-4y\dfrac{dy}{dx}=0$

$\dfrac{dy}{dx}(5x-4y)=-3x^2-5y$

$\dfrac{dy}{dx}=\dfrac{-3x^2-5y}{5x-4y}$ (3)

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2.4.2

$$\frac{dy}{dx}=\frac{-3(1)^2-5(2)}{5(1)-4(2)}=\frac{13}{3}$$

$$y=\frac{13}{3}x+c$$

$$2=\frac{13}{3}(1)+c$$

$$c=-\frac{7}{3}$$

$$y=\frac{13}{3}x-\frac{7}{3}$$ (3)

[24]

QUESTION 3

3.1 3.1.1

$$f(x)=x^3-2x^2+4$$

$$f'(x)=3x^2-4x=0$$

$$x(3x-4)=0$$

$$x=0 \quad \vee \quad x=\frac{4}{3}$$

$$y=4 \qquad y=\frac{76}{27} \quad \vee \quad 2{,}815$$

Thus, the coordinates of the turning points are $(0;4)$ and $\left(\frac{4}{3};\frac{76}{27}\right)$ (3)

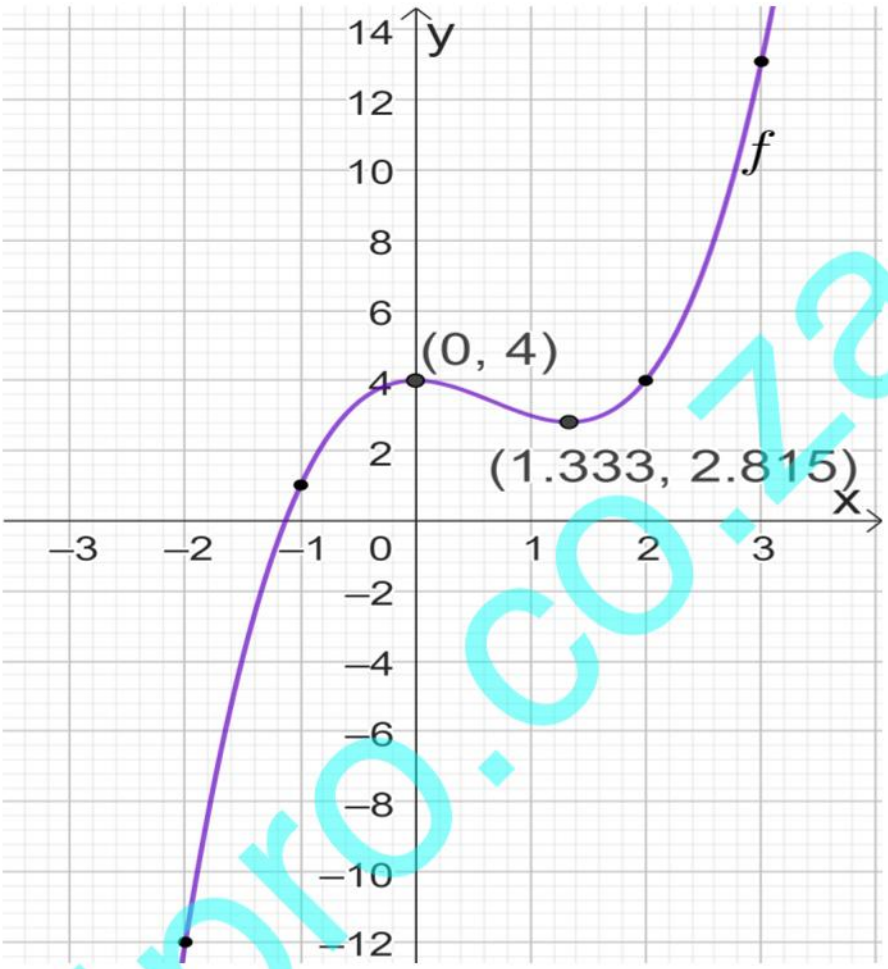
3.1.2

x	-2	-1	0	1	2	3
y	-12	1	4	4	4	14

= 1 mark for every 3 correct answers (2)

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3.1.3



=1 mark for the shape
=1 mark for the indication of the turnings point on the graph (2)

3.1.4 *Let* $x_0 = -1,2$

$f(-1,2) = -0,608$

$f'(-1,2) = 9,120$

$x_1 = -1,2 + \frac{0,608}{9,12}$

$= -1,133$

$f(-1,133) = -0,021$

$f'(-1,133) = 8,383$

(4)

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$$x_2 = -1,133 + \frac{0,021}{8,383}$$

$$= -1,130$$

3.2 Let x = the sides of the square end of the box
y = the length of the box

$$P = 8x + 4y$$

$$144 = 8x + 4y$$

$$y = 36 - 2x$$

$$V = x^2 y$$

$$V = x^2(36 - 2x)$$

$$V = 36x^2 - 2x^3$$

$$\frac{dV}{dx} = 72x - 6x^2 = 0$$

$$6x(12 - x) = 0$$

$$x = 0 \quad \vee \quad x = 12$$
$$y = 36 \quad \vee \quad y = 12$$

Thus, for maximum volume $x = 12$ and $y = 12$. (5)

3.3 $$A = 4\pi r^2$$

$$= 4\pi\left(\frac{D}{2}\right)^2$$

$$= \pi D^2$$

$$\frac{dA}{dt} = 2\pi D \frac{dD}{dt}$$

$$-6\,m^2/hr = 2\pi(4\,m)\frac{dD}{dt}$$

$$\frac{dD}{dt} = \frac{-6\,m^2/hr}{2\pi(4\,m)} = -\frac{3}{4\pi}\,m/hr \quad \vee \; -0,239\,m/hr$$

(4)
[20]

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QUESTION 4

4.1 4.1.1 $\int \frac{2x-1}{(2x^2-2x+5)^2}dx$

$let\ \ u=2x^2-2x+5$

$du=2(2x-1)dx$

$=\int \frac{1}{2}\times\frac{1}{u^2}du$

$=-\frac{1}{2u}+C$

$=-\frac{1}{2(2x^2-2x+5)}+C$ (3)

4.1.2 $\int \sin^5 x\cos^3 x\,dx$

$=\int \sin^4 x\cos^3 x\sin x\,dx$

$=\int (\sin^2 x)^2\cos^3 x\sin x\,dx$

$=\int (1-\cos^2 x)^2\cos^3 x\sin x\,dx$

$=\int (1-2\cos^2 x+\cos^4 x)\cos^3 x\sin x\,dx$

$=\int (\cos^3 x-2\cos^5 x+\cos^7 x)\sin x\,dx$

$let\ \ u=\cos x$
$-du=\sin x\,dx$

$=-\int (u^3-2u^5+u^7)du$

$=-\left[\frac{1}{4}u^4-\frac{1}{3}u^6+\frac{1}{8}u^8\right]+C$

$=-\frac{1}{4}\cos^4 x-\frac{1}{3}\cos x+\frac{1}{8}\cos^8 x+C$

OR

$\int \sin^5 x\cos^3 x\,dx$
$=\int \sin^5 x\cos^2 x\cos x\,dx$

$=\int \sin^5 x(1-sin^2 x)cosx\,dx$

$let\ \ u=\sin x$
$du=\cos x\,dx$

$=\int (u^5-u^7)du$

$=\left[\frac{1}{6}u^6-\frac{1}{8}u^8\right]+C$

$=\frac{1}{6}\sin^6 x-\frac{1}{8}\sin^8 x+C$ (5)

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4.1.3 $\int \frac{2x^4+x^3+x^2-2}{2x+3}dx$

$=\int\left(x^3-x^2+2x-3+\frac{7}{2x+3}\right)dx$

$=\frac{1}{4}x^4-\frac{1}{3}x^3+x^2-3x+\frac{7}{2}\ln(2x+3)+C$ (6)

4.1.4 $\int x\log_3 4x\,dx$

let $u=\log_3 4x$	$dv=xdx$
$du=\frac{1}{x\ln 3}dx$	$v=\frac{x^2}{2}$

$\int \log_3 4x\,dx=\frac{x^2}{2}\log_3 4x-\int \frac{1}{\ln 3}\times\frac{1}{x}\times\frac{x^2}{2}dx$

$=x\log_2 x-\frac{1}{2\ln 3}x^2+C$ (4)

4.2 $\int \frac{2x+7}{x^2+x-2}dx$

$\frac{2x+7}{(x+2)(x-1)}=\frac{A}{x+2}+\frac{B}{x-1}$

$2x+7=A(x-1)+B(x+2)$

let $x=1;$	$B=3$
$x=-2;$	$A=-1$

$\int\left(\frac{-1}{x+2}+\frac{3}{x-1}\right)dx$

$=-\ln(x+2)+3\ln(x-1)+C$ (5)
[23]

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QUESTION 5

5.1 $$t=\int_{10}^{100}\frac{1}{v(2v+1)}dv$$

$$\frac{1}{v(2v+1)}=\frac{A}{v}+\frac{B}{2v+1}$$

$$1=A(2v+1)+Bv$$

---

let $v=0;$ $A=1$

$\vee$ ¿

$v=-\frac{1}{2};$ $A=-2$

$$=\int\left(\frac{1}{v}-\frac{2}{2v+1}\right)dv$$

$$=\left[\ln v-\ln(2v+1)\right]_{10}^{100}$$

$$=(\ln 100-\ln 201)-(\ln 10-\ln 21)$$

$$=0{,}044$$ (5)

5.2 5.2.1 $$x^2=2-x$$

$$x^2+x-2=0$$

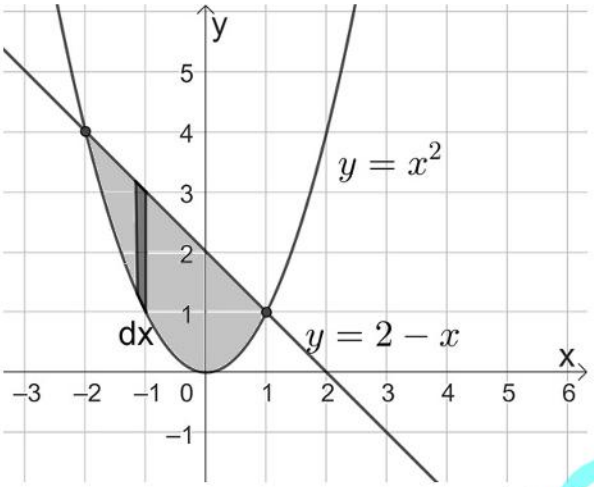
$$(x-1)(x+2)=0$$

$x=1$ $\vee$ $x=-2$
$y=1$ $y=4$

Thus, the coordinates of the points of intersection for bounded area are $(1;1)$ and $(-2;4)$. (3)

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5.2.2



= 1 mark each for shape of the graphs
= 1 mark for indication of the enclosed area and the vertical strip (3)

5.2.3

$$A = \int_a^b (y_1 - y_2)\,dx$$

$$= \int_{-2}^{1} (2 - x - x^2)\,dx$$

$$= \left[2x - \frac{1}{2}x^2 - \frac{1}{3}x^3\right]_{-2}^{1}$$

$$= 4{,}5 \ units^2$$ (3)

5.3

$$dI_z = r^2\,dm$$

$$dm = \frac{m}{A} \times dA = \frac{50}{\pi \times 1{,}5^2} \times 2\pi r\,dr$$

$$dI_z = \frac{400}{9} r^3\,dr$$

$$I_z = \frac{400}{9} \int_0^{1{,}5} r^3\,dr$$

$$= \frac{400}{9}\left[\frac{1}{4}r^4\right]_0^{1{,}5}$$

$$= \frac{100}{9}[1{,}5^2 - 0]$$

$$= 56{,}25 \ kg.m^2$$ (5)

[19]

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QUESTION 6

6.1 $$\frac{dy}{dx}=\frac{(x-1)y^5}{x^2(2y^3-y)}$$

$$\frac{2y^3-y}{y^5}dy=\frac{x-1}{x^2}dx$$

$$-\frac{2}{y}+\frac{1}{3y^3}=\ln x+\frac{1}{x}+C$$ (3)

6.2 $$\frac{d^2y}{dx^2}-9x-12\sin 3x=0, \quad \textit{with} \quad \frac{dy}{dx}=2 \text{ and } y=1 \textit{ at } x=0$$

$$\frac{d^2y}{dx^2}=9x+12\sin 3x$$

$$\frac{dy}{dx}=\frac{9}{2}x^2-4\cos 3x+A$$

$$2=\frac{9}{2}(0)^2-4\cos 3(0)+A$$

$$A=6$$

$$\frac{dy}{dx}=\frac{9}{2}x^2-4\cos 3x+6$$

$$y=\frac{3}{2}x^3-\frac{4}{3}\sin 3x+6x+B$$

$$1=\frac{3}{2}(0)^3-\frac{4}{3}\sin 3(0)+0+B$$

$$B=1$$

$$y=\frac{3}{2}x^3-\frac{4}{3}\sin 3x+6x+1$$ (5)

[8]

TOTAL 100

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